

Mark schemes

Q1.

(a) (i) $\frac{d}{du} \left(2u\sqrt{1+4u^2} \right) = \frac{8u^2}{\sqrt{1+4u^2}} + 2\sqrt{1+4u^2}$

*M1 for clear use of product rule
(condone one error in one term)*

M1

correct unsimplified

A1

$$\frac{d}{du} (\sinh^{-1} 2u) = \frac{2}{\sqrt{1+4u^2}}$$

B1

$$\frac{8u^2 + 2}{\sqrt{1+4u^2}} = \frac{2(1+4u^2)}{\sqrt{1+4u^2}} = 2\sqrt{1+4u^2}$$

be convinced – must see this line OE

$$\frac{d}{du} (2u\sqrt{1+4u^2} + 4\sinh^{-1} 2u) = 4\sqrt{1+4u^2}$$

all working must be correct (not enough to just say k)

Alcso

4

(ii) $\frac{1}{k} \left[2u\sqrt{1+4u^2} + \sinh^{-1} 2u \right]_0^1$

anti differentiation

M1

$$= \frac{\sqrt{5}}{2} + \frac{1}{4} \sinh^{-1} 2$$

FT “their” k or even use of k

A1✓

2

(b) (i) $y = \frac{1}{2} \cos 4x$ and $\frac{dy}{dx} = -2\sin 4x$

substituted into $\int K y \left(1 + \left(\frac{dy}{dx} \right)^2 \right) (dx)$
clear attempt to use formula for CSA

M1

$$(S =) \int_0^{\frac{\pi}{8}} 2\pi \times \frac{1}{2} \cos 4x \sqrt{1+4\sin^2 4x} \, dx = \text{printed answer}$$

(combining $2 \times \frac{1}{2}$)

AG $\frac{dy}{dx} = -2\sin 4x$ **and** $2 \times \frac{1}{2}$ **and** dx must
be seen to award A1cso

A1cso

2

(ii) $u = \sin 4x \Rightarrow du = 4\cos 4x \, dx$
condone $du = B\cos 4x \, dx$ for M1

M1

$$(S =) \frac{\pi}{4} \int_0^1 \sqrt{1+4u^2} \, (du)$$

condone limits seen later

A1

use of their result from (a)(ii) **correctly** FT "their" B

m1

$$(S =) \frac{\pi\sqrt{5}}{8} + \frac{\pi}{16} \sinh^{-1} 2$$

OE

A1cso

4

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Q2.

(a) Use of $\cosh 2x = 2\cosh^2 x - 1$
or $\cosh 4x = 2\cosh^2 2x - 1$

M1

$$\text{RHS} = \frac{1}{2} \cosh 2x + \frac{1}{2} \cosh^2 2x$$

A1

$$= \frac{1}{4} (1 + 2\cosh 2x + \cosh 4x)$$

A1

3

If substituted for both $\cosh 4x$ and $\cosh 2x$ in LHS M1 only, until corrected



If RHS is put in terms of e

M1 for correct substitution

A1 for correct expansion

A1 for correct result

(b) $\frac{dy}{dx} = 2\cosh x \sinh x = \sinh 2x$

allow A1 for $1 + \left(\frac{dy}{dx}\right)^2 = 1 - 4\cosh^2 x + 4\cosh^4 x$

Incorrect form for $\cosh^2 x$ in terms of $\cosh 2x$ M1 only

M1A1

Or

$$y = \left(\frac{e^x + e^{-x}}{2} \right)^2 = \frac{e^{2x} + 2 + e^{-2x}}{4}$$

$$\frac{dy}{dx} = \frac{2e^{2x} - 2e^{-2x}}{4}$$

(M1)

$$= \sinh 2x$$

(A1)

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \sinh^2 2x = \cosh^2 2x$$

AG

A1

3

(c) $S = 2\pi \int_{(0)}^{(\ln 2)} \cosh^2 x \cosh 2x dx$
allow even if limits missing

M1A1

$$= 2\pi \int_0^{\ln 2} \frac{1}{4} (1 + 2\cosh 2x + \cosh 4x) dx$$

m1

$$= \frac{2\pi}{4} \left[x + \frac{2\sinh 2x}{2} + \frac{\sinh 4x}{4} \right]$$

Integrated correctly

A1

Correct use of limits $a = 128$, $b = 495$

accept correct answers written down with no working.

Only one A1 if 2π not used

Q3.

(a) $2y \frac{dy}{dx} = 2x$

Or $\frac{dy}{dx} = x(x^2 + 8)^{\frac{1}{2}}$

B1

$$S = 2\pi \int_0^6 y \sqrt{1 + \frac{x^2}{y^2}} dx$$

M1 for use of formula provided $\frac{dy}{dx}$ is a function of x

M1

A1 for substitution for $\frac{dy}{dx}$ (one slip)

A1F

Eliminating all y

m1

$$= 2\sqrt{2}\pi \int_0^6 \sqrt{x^2 + 4} dx$$

AG

A1

5

(b) $dx = 2 \cosh \theta d\theta$ or $\frac{dx}{d\theta} = 2 \cosh \theta$

B1

$$S = 2\sqrt{2}\pi \int \sqrt{4 \sinh^2 \theta + 4} \cdot 2 \cosh \theta d\theta$$

For eliminating x completely and use of $d\theta$,
ie $d\theta$ attempted

M1

$$S = (2\sqrt{2})\pi \int 2 \cosh \theta \cdot 2 \cosh \theta d\theta$$

Use of $\cosh^2 \theta - \sinh^2 \theta = 1$



m1

$$= 4\sqrt{2}\pi \int (\cosh 2\theta + 1) d\theta$$

Use of formula for $\cosh 2\theta$; must be correct

m1

$$= 4\sqrt{2}\pi \left[\frac{\sinh^2 \theta}{2} + \theta \right]$$

Correct integration of $a \cosh 2\theta + b$

B1F

$$= 4\sqrt{2}\pi [\sinh \theta \cosh \theta + \theta]$$

Use of $\sinh 2\theta = 2 \sinh \theta \cosh \theta$

Must be seen

m1

$$= 4\sqrt{2}\pi \left[\frac{x}{2} \sqrt{\frac{x^2}{4} + 1} + \sinh^{-1} \frac{x}{2} \right]_0^6$$

Or change limits

M1

$$= \pi [24\sqrt{5} + 4\sqrt{2} \sinh^{-1} 3]$$

AG

A1

8

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Q4.

(a) $\frac{dx}{dt} = \sinh 2t$

B1

$$\frac{dy}{dt} = 2 \cosh t$$

B1

$$\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 = \sinh^2 2t + 4 \cosh^2 t$$

M1

Use of $\sinh 2t = 2 \sinh t \cosh t$

Or other correct formula for double angle

m1

$$= 4 \cosh^2 t (\sinh^2 t + 1)$$

For taking out factor

A1

$$= 4 \cosh^4 t$$

ft errors of sign in $\frac{dx}{dt}$ or $\frac{dy}{dt}$

A1F

6

(b) (i) $S = 2\pi \int_0^1 2 \sinh t \cdot 2 \cosh^2 t \, dt$

Using the value obtained in (a)

M1

$$= 8\pi \int_0^1 \sinh t \cdot \cosh^2 t \, dt$$

AG

A1

2

(ii) $S = 8\pi \left[\frac{\cosh^3 t}{3} \right]_0^1$

M1

$$= \frac{8\pi}{3} [\cosh^3 1 - 1]$$

OE eg $\frac{\pi}{3} \left(\left(e + \frac{1}{e} \right)^3 - 8 \right)$

A1

2

[10]

Q5.

(a) $(e^x + e^{-x})^2$ expanded correctly

$e^{2x} + 2e^0 + e^{-2x}$ is acceptable

B1

Result

AG

B1

2

(b) (i) $\frac{dy}{dx} = \sinh x$

B1

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \sinh^2 x}$$

$$= \cosh x$$

$$\text{use of } \cosh^2 x - \sinh^2 x = 1$$

M1

$$S = 2\pi \int_0^{\ln a} \cosh^2 x \, dx$$

AG (clearly derived)

A1

3

(ii) Use of $\cosh^2 x = \frac{1}{2}(1 + \cosh 2x)$
allow one slip in formula

M0 if $\int \cosh^2 x \, dx$ is given as $\sinh 2x$

$$S = \pi \left[x + \frac{1}{2} \sinh 2x \right]_0^{\ln a}$$

A1

$$= \pi \left[\ln a + \frac{1}{2} \left(\frac{e^{2 \ln a} - e^{-2 \ln a}}{2} \right) \right]$$

M1

$$= \pi \left[\ln a + \frac{1}{4} (a^2 - a^{-2}) \right]$$

A1F

$$= \pi \left[\ln a + \frac{1}{4a^2} (a^4 - 1) \right]$$



A1

5

[10]

Q6.

(a) (i) $\frac{d}{dt} \left(\frac{1}{\cosh t} \right) = -1(\cosh t)^{-2} \sinh t$
 $\frac{-2(e^t - e^{-t})}{(e^t + e^{-t})^2}$
 Or

M1A1

$= -\operatorname{sech} t \tanh t$
 AG

A1

3

(ii) Use of $\tanh^2 t = 1 - \operatorname{sech}^2 t$

M1

Printed result

A1

2

(b) (i) $\dot{x} = 1 - \operatorname{sech}^2 t$ ($\dot{y} = -\operatorname{sech} t \tanh t$)

B1

$\dot{x}^2 + \dot{y}^2 = (1 - \operatorname{sech}^2 t)^2 + \operatorname{sech}^2 t - \operatorname{sech}^4 t$
 Any form

M1A1

$= 1 - \operatorname{sech}^2 t = \tanh^2 t$
 AG

A1

4

(ii) $s = \int_0^t \tanh t \, dt$

Ignore limits for M1 and first A1

M1

$= [\ln \cosh t]_0^t$

A1

$$= \ln \cosh t$$

AG

A1

3

(iii) $e^s \cosh t$

M1

$$y = e^{-s}$$

AG

A1

2

(c) $S = 2\pi \int_0^t \operatorname{sech} t \tanh t \, dt$

Ignore limits for M1 and first A1

M1

$$= 2\pi [-\operatorname{sech} t]_0^t$$

A1

$$= 2\pi (1 - \operatorname{sech} t)$$

A1

$$= 2\pi (1 - e^{-t})$$

AG

A1

4

[18]

Q7.

(a) $\dot{x} = 1 - t^2, \dot{y} = 2t$

B1

$$\dot{x}^2 + \dot{y}^2 = (1 - t^2)^2 + 4t^2$$

M1

$$= (1 + t^2)^2$$

AG; must be intermediate line

A1

(b) $S = 2\pi \int_1^2 (1+t^2)t^2 dt$

Must be correct substitutions for M1

M1A1

$$= 2\pi \left[\frac{t^3}{3} + \frac{t^5}{5} \right]_1^2$$

Allow if one term integrated correctly

m1

$$= 2\pi \left[\frac{8}{3} + \frac{32}{5} - \frac{1}{3} - \frac{1}{5} \right]$$

Any form

A1F

$$= \frac{256\pi}{15}$$

A1F

5

[8]

Examiner reports

Q1.

- (a) (i) The differentiation caused more difficulties than had been anticipated. Many students did not use the product rule on the first expression and those who did rarely had both terms correct. Even the derivative of $\sinh^{-1} 2u$ was done badly with many omitting the "2" in the numerator.
- (ii) Those who did not obtain $k = 4$ were still able to score full marks here if they used their value of k or even " k " in their anti-differentiation.
- (b) (i) Students needed to show the correct value of $\frac{dy}{dx}$ before substituting into the formula for the curved surface area in order to score full marks. Despite the printed answer, quite a few omitted the limits or the dx or failed to show the product $2 \times \frac{1}{2}$ in their working and lost an easy accuracy mark.
- (ii) Once again the integration by substitution caused problems for many who omitted the factor $\frac{\pi}{4}$ or who did not have the correct limits for u . Because of errors in finding the value of k in part (a)(i) only the best students scored full marks here. Some credit was given for using their previous results correctly even though students did not have the correct value of k .

Q2.

There were many approaches to the first part of this question, but few completely correct solutions. Some students expressed both sides of the identity in terms of exponential functions in order to show them equal and these were largely successful. Many others worked with either the RHS or the LHS of the identity, often ending up with $\frac{1}{2} \cosh^2 2x + \frac{1}{2} \cosh 2x$ but were unable to complete it convincingly. Part (b) was generally answered well, although not all recognised the identity $2 \sinh x \cosh x = \sinh 2x$ which in turn led to some lengthy solutions. Part (c) was also answered well, with many completely correct solutions, and errors, when they did occur, were largely errors of sign or arithmetic slips.

Q3.

Provided candidates were able to differentiate correctly either by expressing $\frac{dy}{dx}$ in terms of x and y or in terms of x only, they were pretty much successful in part (a). However, candidates were less successful in part (b). Some failed to substitute for $\frac{dy}{dx}$ in terms of θ and so made virtually no progress; others reached $\int \cosh^2 \theta \, d\theta$ but were unable to perform the integral. Of those who realised that $\cosh^2 \theta$ needed to be expressed in terms of $\cosh 2\theta$, some made a sign error in the double angle formula. However, most of those candidates who managed to correctly integrate were unable to evaluate $\sinh 2(\sinh^{-1} 3)$ and merely wrote down the printed answer, which of course was not acceptable. Few candidates thought of expressing $\sinh 2\theta$ in terms of $\sinh \theta$ and $\cosh \theta$.

Q4.

Although a good number of candidates answered part (a) correctly, quite a few stalled at the handling of $\sinh^2 2t$ either by misquoting a formula for the double angle or by using longwinded algebraic methods in which they lost direction. Consequently these candidates were unable to score full marks in part (b)(i).

Part (b)(ii) proved to be beyond the abilities of the majority of candidates. The usual attempts were either to express $\cosh^2 t$ by $1 + \sinh^2 t$ or to express $\cosh^2 t$ in terms of $\cosh 2t$, thereby making no progress. Few thought of using a simple substitution.

Q5.

Solutions to part (a) were usually correct although not always very elegant. Part (b)(i) was usually correctly worked although $\sqrt{1 + \sinh^2 x} = 1 + \sinh x = \cosh x$ did appear more often than expected. There were some good solutions to part (b)(ii) and when solutions

were incomplete, they almost always ended at $S = \pi \left(\ln a + \frac{1}{2} \sinh(2 \ln a) \right)$. It should perhaps be noted that when answers are printed, candidates must give sufficient detail in their working to show how that answer is arrived at.

Q6.

Generally, candidates scored quite well on this question. Some candidates struggled with part (a)(i) by not realising that $\operatorname{sech} t$ was $(\cosh t)^{-1}$, instead expressing $\operatorname{sech} t$ in exponential form.

This latter method rarely led to a correct solution. However, apart from some sign fudging in part (a)(ii), most candidates were able to recover to answer parts (a)(ii) and (b)(i) correctly. Very few candidates were able to score the three available marks in part (b)(ii), by either ignoring the limits of integration completely or by writing $s = \ln \cosh t + c$ with no effort to show that the value of c was zero. In spite of some inelegant methods, part (b)(iii) was usually answered correctly.

Part(c) caused problems, often by candidates attempting to integrate $e^{-s} \tanh t$ with respect to t by regarding e^{-s} either as a constant or else as e^{-t} . Of those who correctly integrated to arrive at $-\operatorname{sech} t$, again very few candidates recognised the need for limits and so were unable to arrive at the printed result.

Q7.

This was a very well-answered question with the vast majority of candidates either gaining full marks or losing one mark through faulty arithmetic. Very occasionally a candidate

differentiated $t^2 + t^4$ or $t^3 t^4$ integrated $t^2 + t^4$, but wrote down $\frac{t^3}{3} + \frac{t^4}{4}$.